

AETHERBEAM: HIGH-FIDELITY NONLINEAR WAVEGUIDE ACOUSTIC RAY-TRACING ENGINE

Mathematical Formulation and Physical Principles of Orders 0–4 Sound Beam Synthesis

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AetherBeam Reverb DSP Architecture • Advanced Spatial Audio

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ABSTRACT

This paper presents the complete mathematical architecture of the **AetherBeam** physical acoustic engine. The system synthesizes spatial impulse responses by combining higher-order 3D geometric image-source ray tracing (Orders $0 \leq n \leq 4$, producing up to 96 physical acoustic reflection paths per enclosure), analytic boundary collision validation, and finite-amplitude weakly nonlinear acoustic wave propagation (Fubini-Bessel shock wave steepening under extreme sound pressure levels up to 122 dB SPL, representative of a full symphony orchestra playing at *fff*).

1. 3D Architectural Image-Source Geometry (Orders 0–4)

Let the bounding coordinates of an acoustic enclosure $\Omega \subset \mathbb{R}^3$ be denoted by:

$$\Omega = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [z_{\min}, z_{\max}]$$

with enclosure dimensions $L_x = x_{\max} - x_{\min}$, $L_y = y_{\max} - y_{\min}$, and $L_z = z_{\max} - z_{\min}$. Let $\mathbf{s} = (s_x, s_y, s_z)^\top \in \Omega$ denote the physical sound source, and $\mathbf{r} = (r_x, r_y, r_z)^\top \in \Omega$ denote the listener position.

1.1 Lattice Coordinate Transformation

Relative to the enclosure minimum origin $\mathbf{b}_{\min}$, coordinates are mapped to $\mathbf{s}_{\text{rel}} = \mathbf{s} - \mathbf{b}_{\min}$ and $\mathbf{r}_{\text{rel}} = \mathbf{r} - \mathbf{b}_{\min}$. For reflection index vector $\mathbf{n} = (n_x, n_y, n_z) \in \mathbb{Z}^3$, the virtual image-source coordinate along axis $i \in \{x, y, z\}$ is defined by:

$$s'_i(n_i) = \begin{cases} n_i L_i + s_{\text{rel},i}, & \text{if } n_i \equiv 0 \pmod{2} \\ (n_i + 1) L_i - s_{\text{rel},i}, & \text{if } n_i \equiv 1 \pmod{2} \end{cases}$$

The total reflection order $N(\mathbf{n})$ is the ℓ_1 norm of the lattice vector:

$$N(\mathbf{n}) = |n_x| + |n_y| + |n_z|, \quad 0 \leq N(\mathbf{n}) \leq 4$$

1.2 Ray Trajectory and Propagation Metric

The vector from the virtual image-source $\mathbf{s}'(\mathbf{n})$ to the receiver $\mathbf{r}_{\text{rel}}$ is $\mathbf{d}(\mathbf{n}) = \mathbf{r}_{\text{rel}} - \mathbf{s}'(\mathbf{n})$, with total propagation distance:

$$d(\mathbf{n}) = \|\mathbf{d}(\mathbf{n})\|_2 = \sqrt{(r_{\text{rel},x} - s'_x)^2 + (r_{\text{rel},y} - s'_y)^2 + (r_{\text{rel},z} - s'_z)^2}$$

The corresponding acoustic propagation delay in dry air at 20°C ($c_0 = 343.2$ m/s) is:

$$\tau(\mathbf{n}) = \frac{d(\mathbf{n})}{c_0}$$

1.3 Trajectory Folding

The folded spatial position $\mathbf{p}(t)$ along normalized path parameter $t \in [0, 1]$ is obtained via periodic coordinate folding:

$$p_i(t) = \text{fold}(s'_i + t \cdot d_i(\mathbf{n}), L_i) + b_{\min,i}$$

2. Strict Architectural Boundary Containment

To ensure sound rays never escape non-rectangular or vaulted architectures, every reflection point $\mathbf{p} \in \mathbb{R}^3$ must satisfy the space-specific manifold constraint $\mathcal{C}_{\text{space}}(\mathbf{p}) \leq 0$.

2.1 Sydney Opera House (Pitched Vault)

$$\mathcal{C}_{\text{Sydney}}(\mathbf{p}) = p_z - (25.0 - |p_x|) \leq 0$$

Any virtual ray whose folded reflection apex exceeds $Z = 25 - |x|$ is immediately pruned.

2.2 Rome Pantheon (Cylindrical Drum + Hemispherical Dome)

$$\mathcal{C}_{\text{Pantheon}}(\mathbf{p}) = \begin{cases} \sqrt{p_x^2 + p_y^2} - 21.65 \leq 0, & \text{if } p_z \leq 21.65 \text{ m} \\ p_x^2 + p_y^2 + (p_z - 21.65)^2 - (21.65)^2 \leq 0, & \text{if } p_z > 21.65 \text{ m} \end{cases}$$

2.3 Hagia Sophia (Central Dome + Exedrae)

For the central nave within radius $r = \sqrt{p_x^2 + p_y^2} \leq 15.5 \text{ m}$:

$$p_z \leq 25.0 + 30.0 \sqrt{1 - \left(\frac{r}{15.5}\right)^2}$$

3. Finite-Amplitude Nonlinear Wave Steepening (fffDynamics)

When a full symphony orchestra plays fortississimo (*fff*), sound pressure levels near the proscenium reach $L_p \approx 122 \text{ dB SPL}$, corresponding to peak acoustic pressure:

$$p_{\text{peak}} = p_{\text{ref}} \cdot 10^{\frac{L_p}{20}} = (20 \mu\text{Pa}) \cdot 10^{\frac{122}{20}} \approx 25.18 \text{ Pa}$$

3.1 Burgers' Equation and Nonlinearity Parameter

One-dimensional finite-amplitude plane wave propagation is governed by Burgers' equation:

$$\frac{\partial p}{\partial x} - \frac{\beta p}{\rho_0 c_0^3} \frac{\partial p}{\partial t'} = \frac{\delta}{2c_0^3} \frac{\partial^2 p}{\partial t'^2}$$

where $t' = t - x/c_0$ is retarded time, $\rho_0 = 1.204 \text{ kg/m}^3$, $c_0 = 343.2 \text{ m/s}$, and $\beta = 1 + \frac{B}{2A} \approx 1.201$ is the nonlinearity parameter for air.

3.2 Shock Formation Distance and Fubini-Bessel Series

The Goldberg distortion parameter $\sigma(d)$ along path length d is:

$$\sigma(d) = \frac{\beta \omega p_{\text{peak}} d}{\rho_0 c_0^3}$$

For symphonic orchestral dynamics ($\sigma \in [0.05, 0.45]$), the acoustic pressure field obeys the **Fubini-Bessel series**:

$$p(d, t') = p_{\text{peak}} \sum_{n=1}^{\infty} \frac{2}{n\sigma(d)} J_n(n\sigma(d)) \sin(n\omega t')$$

3.3 Discrete Nonlinear Waveguide Implementation

At $f_s = 96$ kHz, cumulative wave steepening along path k of distance d_k is evaluated via polynomial wave-shaping with dynamic coefficient γ_k :

$$\gamma_k = \frac{\beta \cdot p_{\text{peak}} \cdot d_k}{2\rho_0 c_0^3}, \quad y_k[n] = x[n] + \gamma_k \cdot x^2[n] - \frac{1}{3}\gamma_k^2 \cdot x^3[n]$$

4. Total Impulse Response Synthesis

The complete binaural impulse response $h_{L,R}(t)$ is synthesized as the superposition of deterministic early physical beams (Orders $0 \leq n \leq 4$) and the stochastic diffuse late reverberant tail:

$$h_{L,R}(t) = \sum_{k=1}^K A_k \cdot \mathcal{N}_k(\delta(t - \tau_k) * h_{\text{air}}(t, d_k)) \cdot \text{HRTF}_{L,R}(\theta_k, \phi_k) + h_{\text{late}}(t)$$

where $K \leq 96$ is the total count of valid physical rays, $A_k = R_k/d_k$ accounts for spherical attenuation and boundary absorption, $\mathcal{N}_k(\cdot)$ is the nonlinear wave-steepening operator, and $h_{\text{late}}(t) = e^{-3 \ln(10)t/RT_{60}} \cdot \xi(t)$ provides the late reverberation tail.

Table 1: Physical acoustic parameters of synthesized showcase spaces.

Enclosure	Dimensions (m)	Volume (m ³)	RT ₆₀ (s)	Order	Valid Rays	L _p (SPL)
Musikverein Golden Hall	48.8 × 19.1 × 17.7	15,000	2.05	4	76	122 dB (<i>fff</i>)
Sydney Opera House	52.0 × 26.0 × 25.0	16,500	2.00	4	68	122 dB (<i>fff</i>)
Hagia Sophia Byzantine	65.0 × 60.0 × 55.0	180,000	11.00	4	84	122 dB (<i>fff</i>)